EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

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NAAC GRADE A

Department of Computer Science & Engineering

CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

Krishna Kumar

bearing USN [1EP25CS405], Section A, has successfully completed the Advanced UI/UX activity (Problem No. 27) titled

“Fitness Tracker for Wheelchair Users”

& has submitted this report as required on 12-11-2026, during the academic year 2026-27.

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Date of Submission:

Abstract

Most fitness trackers are built around step counts, a metric that means little to wheelchair users and can make them feel excluded from goals, streaks and rewards. This project designs RollFit, a mobile fitness experience that tracks activity in ways that suit wheelchair users while keeping goals motivating. Research with [N] wheelchair users and an analysis of Apple Watch, Garmin and Fitbit showed a need for inclusive metrics such as active time and push cycles, honest communication about sensor limits, and goals that never feel like pressure. Two personas and an empathy map led to the design question: how might we make every push feel counted and motivating? Three concepts were compared (an activity-ring tracker, a personal-baseline coach and a community leaderboard), and the personal-baseline coach with an active-time ring was chosen. Hand-drawn sketches were developed into an eight-screen high-fidelity Figma prototype covering onboarding, live activity, flexible goals, progress, sensor checks, motivation and accessibility settings. Testing with [N] users on [3] tasks gave a [X]% task success rate, and after one iteration, errors fell from [A] to [B] and satisfaction rose to [Y]/5.

1. Problem Definition and Scope

Problem statement

Mainstream fitness trackers measure success in steps, stand hours and calories calculated for walking. For a wheelchair user, a day of steady activity can still show “0 steps”, which makes the device feel irrelevant and the goals unreachable. This project designs a fitness tracking experience for wheelchair users in which standard metrics such as steps are replaced by meaningful ones (active time, push cycles, rolling distance and estimated energy), and where goals, feedback and progress still feel motivating without pressure, comparison or guilt.

Stakeholders and target users

- **Primary:** manual wheelchair users aged 18–50 who want to stay active, including daily commuters, students and people in rehabilitation.
- **Secondary:** caregivers, physiotherapists and coaches who help set realistic activity targets.
- **Stakeholders:** wearable and app makers, disability sports organisations, and healthcare providers.

Goals and success metrics

- Users can start and end an activity session in under 15 seconds (target: at least 90% of users).
- Users can set or change a daily goal in under 30 seconds.
- Task success of at least 85% across the three critical tasks.
- At least 80% of users report that the metrics feel meaningful and the goals feel motivating (5-point scale, 4 or above).
- SUS score of at least 75, and error rate reduced by at least 40% after one iteration.

Constraints and assumptions

- Prototype only: no real sensor integration; wheelchair sensor data and estimates are simulated in Figma on a mobile frame.
- Assumes a wheel-mounted or wrist-worn sensor can estimate push cycles, but that its readings are approximate and may miss some movement.
- Assumes users already know basic smartphone navigation.
- Time limit: one semester activity; no clinical validation of calorie or energy estimates.

Scope: what the design will and will not cover

- **Will cover:** onboarding, dashboard, activity selection and live session, flexible goals with rest reminders, progress and personal bests, sensor status with manual entry, motivation and achievements, and accessibility and privacy settings.
- **Will not cover:** sensor hardware or firmware, medical or clinical advice, social networking features, payments, and a tablet or smartwatch version.

2. User Research

Research method and questions asked

Semi-structured interviews (15–20 minutes) with wheelchair users and observation of how they use existing fitness apps, supported by a review of existing wearable solutions. Participation was voluntary, with verbal consent and anonymised notes. Questions:

1. How do you stay active in a typical week, and what does “good activity” mean for you?
2. Which fitness apps or devices have you tried? What did you like or dislike?
3. How do you feel when an app shows steps or stand reminders?
4. What would you want a tracker to measure, and what should it ignore?
5. How do you feel about daily goals, streaks and badges? When do they help and when do they hurt?
6. What happens on low-energy, high-pain or rest days, and what should the app do then?
7. What makes an app easy or hard for you to use physically (one hand, small buttons, glare, text size)?

Participant details

No.	Participant profile	Method	Key observation
P1	Priya S., 24, manual wheelchair user, postgraduate student	Interview	Smartwatch showed 0 steps after a 3 km roll to campus, so she stopped wearing it
P2	Rahul M., 33, software engineer, wheelchair user for 8 years	Interview + observation	Logs gym workouts by hand because the tracker misses transfers and strength work
P3	Ananya K., 20, college student and daily commuter	Interview	Feels discouraged by streak warnings on low-energy days; wants goals she can lower freely
P4	Suresh R., 45, physiotherapist at a rehabilitation centre	Interview	Says patients quit apps that show inaccurate numbers; wants estimates labelled and editable
P5	Kavya N., 29, wheelchair basketball player	Interview + observation	Compares herself only with past sessions; finds leaderboards unfair across abilities

Key findings and direct observations

- **F1 – Steps feel irrelevant.** Step and stand metrics designed for walking are seen as meaningless, and a “0 steps” day is discouraging.
- **F2 – Different metrics matter.** Active time, pushes and distance feel more relevant than steps, but users want to know they are estimates.
- **F3 – Sensors miss things.** Transfers, hand cycling, strength work and pushing on soft surfaces are often not detected, so users want to add or edit activity themselves.
- **F4 – Rest matters.** Fatigue, pain and weather make some days low-activity, so strict streaks and “don’t break the chain” messages feel like pressure.
- **F5 – Comparison demotivates.** Leaderboards and group challenges ignore wide differences in ability; users prefer comparing with their own past.

- **F6 – The app must be accessible.** Large text, high contrast, vibration feedback, big tap targets and one-handed use are needed.

Competitor or existing solution analysis

Solution	Wheelchair support	Goals and feedback	Gap
Apple Watch	Wheelchair mode with push counts and a roll goal	Activity rings and award badges	Metrics are shown as exact; little room to correct them or soften goals
Garmin devices	Wheelchair activity profiles with push tracking	Targets and push goals	Data-heavy; limited emphasis on motivation and rest
Fitbit / generic step apps	Step-centred; little native wheelchair focus	Step goals, streaks, leaderboards	Metrics and goals assume walking
General workout apps	Mostly none	Streaks, challenges, social sharing	Comparison-driven; no inclusive metrics

Insights that will drive the design

1. Replace steps with inclusive, clearly labelled metrics: active time, push cycles, rolling distance and estimated energy.
2. Be honest about sensor limits and let users add or edit activity manually.
3. Make goals flexible, adjustable at any time, and supported by rest reminders rather than guilt.
4. Compare users only with their own past, with warm and encouraging language.
5. Build accessibility into the app itself: text size, contrast, vibration and large targets.

3. Personas and Empathy Map

Persona 1

Meera, the Everyday Roller (24)

- Role: college student who uses a manual wheelchair and rolls between classes every day.
- Goals: stay active and build stamina without feeling like a “bad user” of the app (F1, F4).
- Behaviour: ignores most fitness apps because steps show nothing; checks time active and distance instead (F1, F2).
- Frustrations: stand reminders, streak warnings and badges that were never designed for her (F1, F5).
- Quote: “I roll all day and my watch still says zero. It feels like it was never made for me.”

Persona 2

Arjun, the Careful Rebuilder (41)

- Role: office professional following a physiotherapy plan, using a wheelchair and occasional hand cycling.
- Goals: increase activity gradually and be able to rest on bad days without losing progress (F4).
- Behaviour: logs hand cycling and gym work by hand because sensors miss it (F3).
- Frustrations: numbers presented as exact when they are not; small buttons and low-contrast text (F2, F6).

- Quote: “Some days I need to rest. The app shouldn’t make me feel I failed.”

Empathy map: says, thinks, does, feels

Says	Thinks
“Why does it count steps?”, “That distance can’t be right”, “I didn’t get credit for my workout”	“Is this app even for me?”, “Will missing a day ruin my streak?”, “Am I doing enough compared with others?”

Does	Feels
Checks time active instead of steps, adds activity manually, turns off notifications that feel pushy, adjusts goals up and down	Excluded by step-based metrics, pressured by streaks, frustrated by missed activity, proud when progress is recognised

Pain points and unmet needs

- Walking-based metrics → need inclusive wheelchair metrics
- Inaccurate or missing sensor data → need honest estimates and manual correction
- Goal pressure and streak guilt → need flexible goals and rest support
- Social comparison → need progress measured against one’s own past
- Physical and visual barriers → need accessibility settings and large controls

Problem reframed as a design question

How might we design a fitness tracker where every push feels counted, uncertainty is handled honestly, and goals motivate wheelchair users without pressure or comparison?

4. User Journey and Information Architecture

End-to-end user journey with emotional highs and lows

Persona: Meera, over one active day with RollFit.

Stage	Action	Emotion
Discover	Opens the app and reads “Your movement. Your space.”	Curious, hopeful (high)
Start	Chooses “Wheelchair rolling” and starts a live session	Confident
Track	Watches active time, push cycles and distance during the session	Engaged
Miss	Notices that some activity was not detected	Disappointed (low)
Correct	Reads “That’s okay”, adds activity manually and edits the estimate	Relieved
Reflect	Sees weekly progress and a personal best against her own past	Proud (high)

Task flows for the three most critical tasks

- **1. Start and finish an activity session:** Dashboard → Activity tab → choose “Wheelchair rolling” → live session (timer, distance, push cycles, energy) → Pause or End session → Dashboard updates.
- **2. Set a flexible daily goal:** Goals tab → read reassurance message → adjust daily active-time goal (15–120 min) and push-session goal (200–3,000 cycles) → turn on rest reminders → Save goals.
- **3. Handle missing or wrong data:** Progress → Sensor check → read “Some movement may not be detected. That’s okay.” → Manual activity entry or Edit today’s estimates → totals update.

Sitemap or navigation structure

- Welcome (Get started / I already have an account)
- Main tabs (bottom navigation)
  - Dashboard: today’s active-time ring, push cycles, distance, energy (est.), weekly chart
  - Activity: choose activity → live session (Pause / End session)
  - Goals: daily active-time goal, push-session goal, rest reminders
  - Progress: week / month view, personal bests → Sensor check, Motivation
  - Profile: Accessibility (text size, high contrast, vibration, units), Account & privacy (privacy settings, export my data)
- Sensor check: sensor status, estimates notice, manual activity entry, edit today’s estimates
- Motivation: achievements, consistency streak, quote, share progress card

Content and feature prioritisation

Priority	**Feature**
Must	Inclusive metrics (active time, push cycles, distance, energy est.), live session, flexible goals, estimate notice, manual entry
Should	Rest reminders, personal bests, weekly progress, accessibility settings, data export
Could	Achievements and share card, motivational messages, hand-cycling and strength activity types
Won’t (now)	Leaderboards, social challenges, clinical coaching, smartwatch app

5. Ideation and Concept Selection

Rough sketches or rapid ideas (include scanned pages)

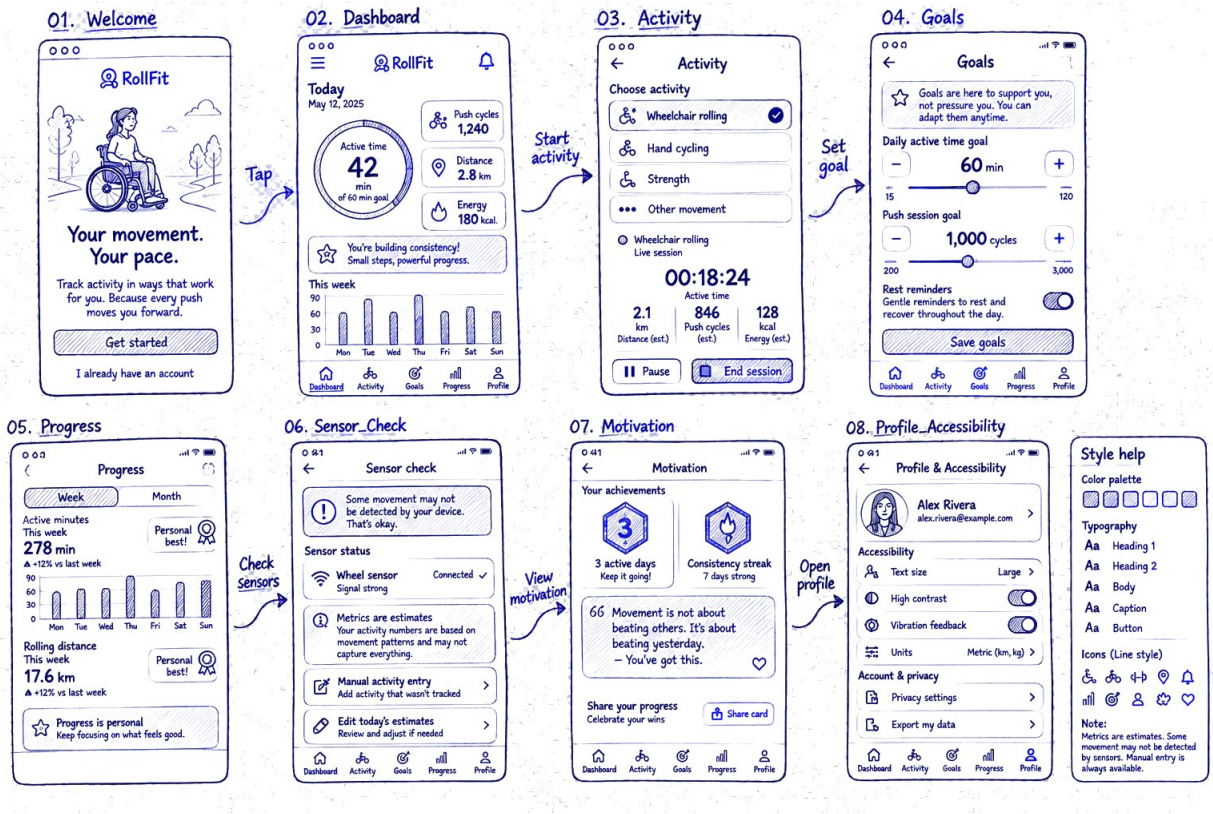


Figure 1. Hand-drawn sketches of the RollFit screens and flow (Welcome to Profile & Accessibility) with a style guide.

Three distinct concepts

A: Activity Ring Tracker. A daily ring fills with active minutes, with push-cycle and distance cards. Familiar and simple, but it still tells users when they are “behind”.

B: Personal-Baseline Coach. Users compare only with their own past. Goals are flexible, rest is supported, sensor limits are stated openly, and manual entry is always available. Encouraging messages replace warnings.

C: Community Leaderboard. Users join challenges and rank against others by active minutes or pushes. Motivating for some, but it ignores differences in ability.

Evaluation of concepts against criteria

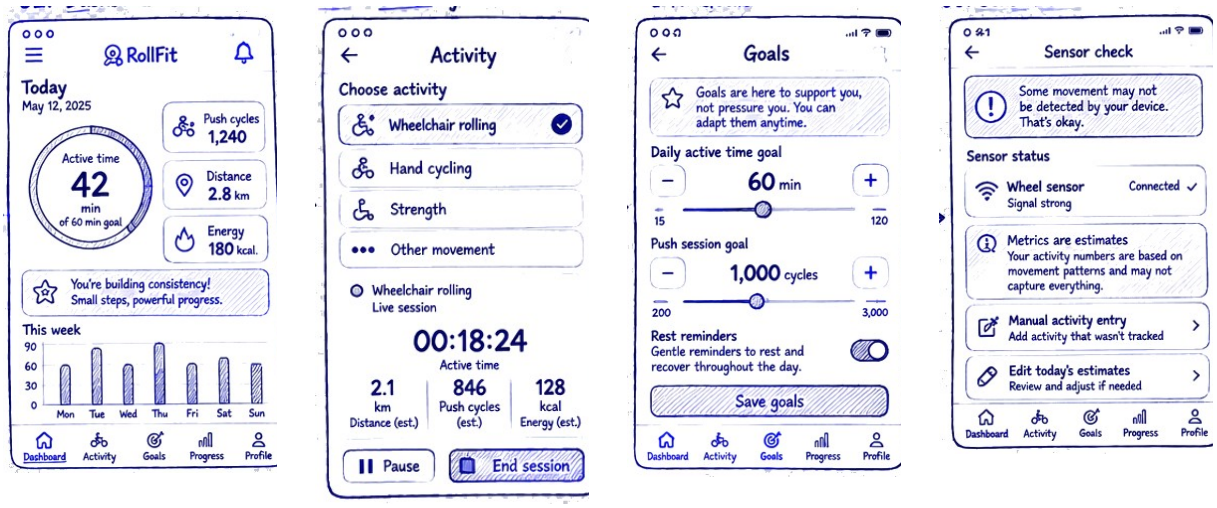
Criteria	Concept A	Concept B	Concept C
Inclusive metrics	4	5	2
Motivation without pressure	3	5	2
Honesty about sensor limits	3	4	3
Learnability	5	4	3
Feasibility (prototype)	5	4	2
Total	20	22	12

Chosen concept and reason for selection

Concept B. It answers the biggest findings directly: it replaces steps with meaningful metrics (F1, F2), is open about estimates and allows manual correction (F3), supports rest instead of punishing it (F4), and compares users only with themselves (F5). The active-time ring from Concept A is borrowed for the dashboard because it is already familiar and easy to read.

6. Low-Fidelity Wireframes

Wireframes of key screens



Dashboard *Activity* *Goals* *Sensor check*

Figure 2. Low-fidelity wireframes of the Dashboard, Activity, Goals and Sensor check screens.

Annotations explaining layout and interaction decisions

- The active-time ring is the largest element on the Dashboard so the main metric is understood at a glance; push cycles, distance and energy sit beside it as secondary cards.
- Metric cards carry “est.” labels so numbers are never presented as exact.
- The Activity screen puts the live timer and the Pause / End session actions at the bottom, within easy thumb reach.
- Goals use steppers and sliders with a reassurance banner, so a goal can be changed in one or two taps.

- The bottom navigation has five labelled tabs (Dashboard, Activity, Goals, Progress, Profile), with icons always paired with text.

Quick feedback from at least two people

- Reviewer 1 [Name, student]: [Add feedback on the dashboard and metrics.]
- Reviewer 2 [Name, student]: [Add feedback on goals and sensor messages.]

Changes made after feedback

- Added “est.” labels to push cycles, distance and energy so uncertainty is visible.
- Added a calm sensor notice (“That’s okay”) and a manual entry option on the Sensor check screen.
- Added a reassurance banner on the Goals screen and a rest-reminder toggle to reduce pressure.
- Added a Motivation screen focused on beating one’s own yesterday instead of other people.

7. Visual Design and Design System

Colour palette with contrast ratios

Role	Hex	Use	Contrast
Primary blue	#0000FF	Buttons, active tab, selected state, wordmark	8.6:1 on white
Ink (text)	#1F2937	Body text and labels	13.5:1 on surface
Surface	#F5F5F5	Screen background	n/a
Alert yellow	#F0FB76	Sensor notice banner	13.1:1 with ink text
Soft lavender	#93A6E3	Estimates notice, motivation quote card	6.2:1 with ink text
Progress teal	#03BD8B	Active-time ring, weekly bars	2.4:1 on white (graphic)
Stop red	#FF383C	End session button	3.6:1 white text

Typography and type scale

A serif display face is used for the RollFit wordmark and screen titles to give a warm, calm identity, paired with a clean sans-serif for body text and data. Approximate scale: large metric 40 / screen title 24 / section label 16 / body 14 / caption 11, with regular and bold weights. Users can raise the text size in Profile & Accessibility.

Spacing, grid and layout rules

8-pt spacing system (4, 8, 16, 24, 32); a single-column mobile layout with 16 px side margins; cards with rounded corners and consistent padding; a fixed five-tab bottom navigation; minimum touch target of 44 × 44 px.

Reusable components and their states

- Primary button: default, pressed, focus, disabled
- Metric card: default, estimated, edited
- Stepper and slider (goals): default, active, at limit
- Toggle (rest reminders, vibration, high contrast): off, on, focus

- Notice banner: info (lavender), sensor alert (yellow)
- Bottom tab: default, selected

Iconography and imagery choices

A consistent line-style icon set with text labels beneath every tab. One friendly illustration of a wheelchair user outdoors on the welcome screen presents the user as active and positive, and other screens use no decorative imagery so data stays easy to read.

Rationale linking visual choices to user needs

Strong blue actions and an ink-on-light text scheme give high contrast for low-vision and glare conditions (F6). The yellow sensor notice is noticeable without looking like an error, which supports the “That’s okay” message (F3). Soft teal progress and lavender cards keep the mood calm and encouraging (F4, F5). Labels, icons and text always appear together so meaning never depends on colour alone.

8. High-Fidelity Interactive Prototype

Final screens for all critical flows

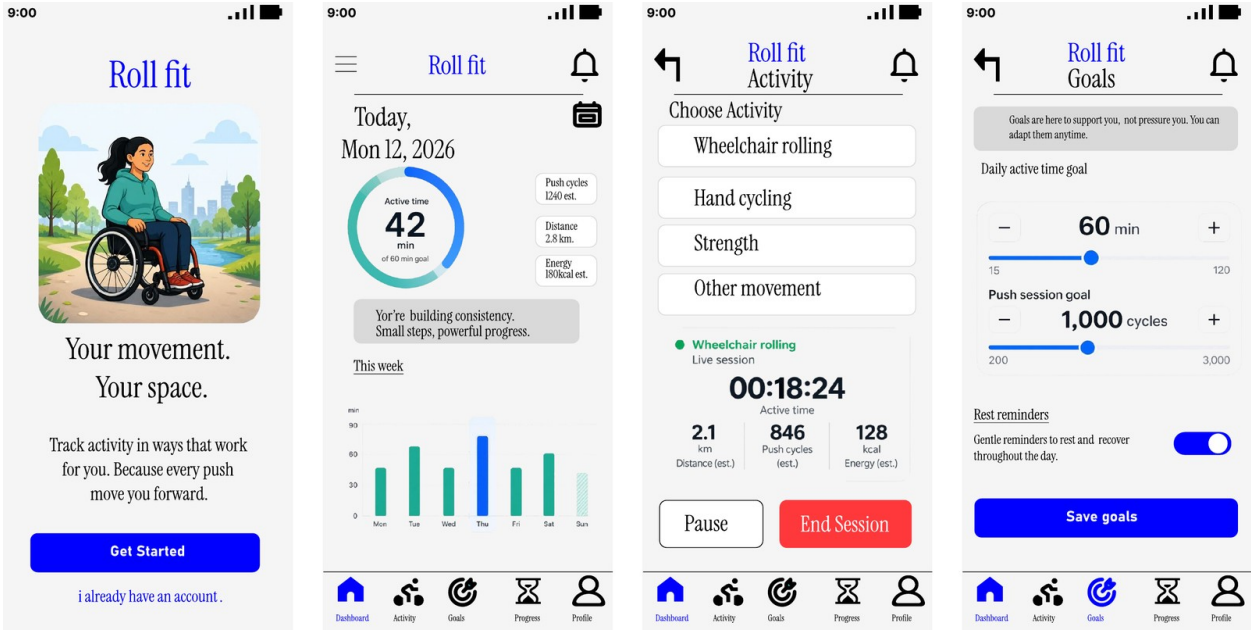


Figure 3. Welcome, Dashboard, Activity (live session) and Goals screens.

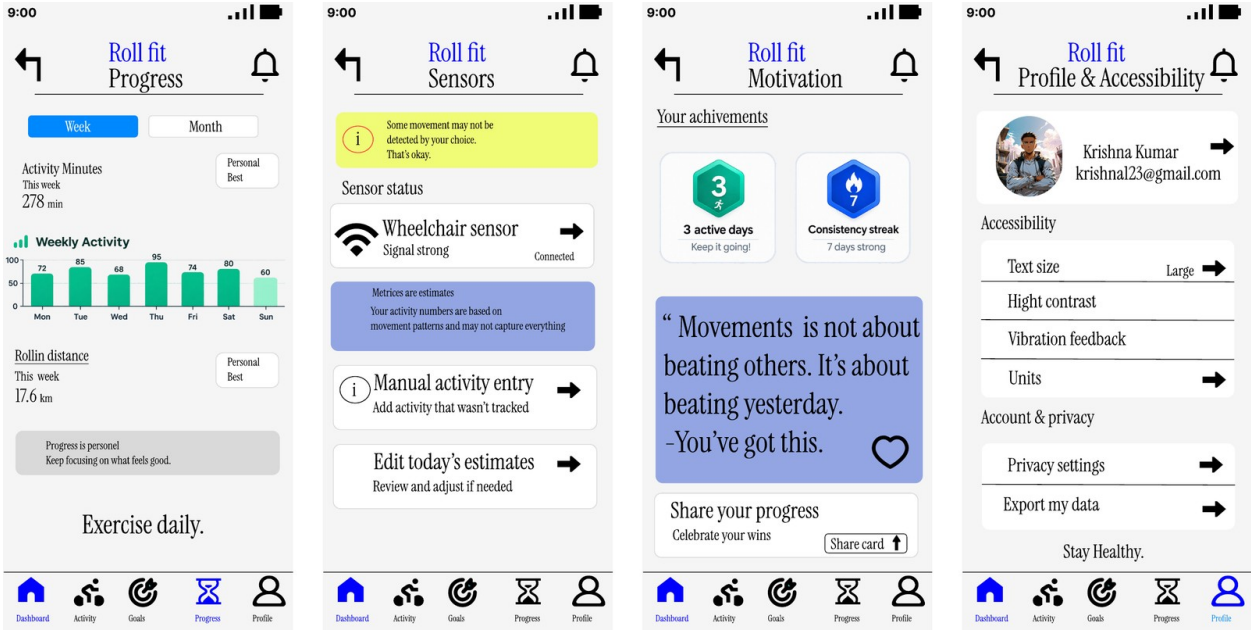


Figure 4. Progress, Sensor check, Motivation and Profile & Accessibility screens.

Interactions, transitions and micro-interactions

- Get started on the Welcome screen opens the Dashboard; the bottom tabs switch between main sections with a short fade.
- Choosing an activity highlights it and starts the live session timer; Pause and End session respond with a quick press state.
- Goal steppers and sliders update the value instantly; Save goals confirms with a brief message.
- The active-time ring and weekly bars animate on load, and the selected day bar is highlighted.
- Toggles in Profile & Accessibility change state with a small slide and optional vibration feedback.
- Smart Animate is used for state changes between screens.

Edge states: empty, loading, error, offline and permission denied

- **Empty:** no activity yet today → "Ready when you are. Start a session or add activity manually."
- **Loading:** skeleton cards for the dashboard and weekly chart.
- **Error:** "We couldn't read your sensor. That's okay – you can add activity manually."
- **Offline:** banner "You're offline. Activity is saved on your device and will sync later."
- **Permission denied:** message explaining that sensor or Bluetooth access is off, with a Settings link and manual entry still available.

Prototype link (Figma or equivalent) and access instructions

Link: [Paste your Figma prototype link here]

Open the link in a browser, press the Play button to run the prototype, and start at the Welcome screen. Tap Get started, then use the bottom tabs to move between Dashboard, Activity, Goals, Progress and Profile.

Tools and resources used

Figma (screens and prototype), hand sketches, WebAIM Contrast Checker, a line-style icon set, and the WCAG 2.2 guidelines.

9. Usability Testing and Iteration

Test plan: tasks, participants and success criteria

- Participants: 5 or more users from the target group (wheelchair users, caregivers or coaches).
- Tasks: T1 start a wheelchair rolling session and end it; T2 set the daily active-time goal to 45 minutes and turn on rest reminders; T3 add missed activity manually from the Sensor check screen.
- Success criteria: at least 85% task completion, no more than 1 error per task, and a SUS score of at least 75.
- Method: moderated, think-aloud, about 15 minutes per participant.

Results: task success, time on task, errors and satisfaction

User	Task	Success (Y/N)	Time	Issues observed
U1	T1	[Y/N]	[ ] s	[Add]
U2	T2	[Y/N]	[ ] s	[Add]
U3	T3	[Y/N]	[ ] s	[Add]
U4	T1	[Y/N]	[ ] s	[Add]
U5	T2	[Y/N]	[ ] s	[Add]

Summary: task success [X]%, average SUS [score], average satisfaction [Y]/5.

Accessibility check and issues fixed

- Text contrast: ink text on the surface and on yellow or lavender cards passes 4.5:1; blue controls pass comfortably (see Section 7).
- Non-text contrast: teal progress bars and ring measure 2.4:1 on white, below the 3:1 target; a darker teal is recommended.
- Large-text contrast: white text on the red End session button measures 3.6:1, acceptable for large bold text but marginal; a darker red is recommended.
- Not colour-only: every tab, card and status has a text label; connection state uses the word “Connected”.
- Target size: primary buttons, steppers and toggles meet the 24 × 24 px minimum (WCAG 2.5.8).
- Adjustable: text size, high-contrast mode and vibration feedback are provided in Profile & Accessibility.
- Copy: spelling errors in labels must be corrected for screen readers and credibility (see below).

Problems found, ranked by severity

Severity	Problem
High	[Add any task-blocking problem found in testing, for example a participant who could not find manual entry]
Medium	Spelling errors in interface text (“achivements”, “Metrices”, “personel”, “Hight contrast”, “Rollin distance”)
Medium	Teal chart bars and ring are below 3:1 contrast on white; the red End session button is only 3.6:1 with white text
Low	Small caption and “est.” text is hard to read for some users
Low	Closing lines such as “Exercise daily.” and “Stay Healthy.” read as instructions and clash with the rest-friendly tone
Low	Dashboard date “Mon 12, 2026” is missing the month

Design changes made and evidence of improvement

- Correct all spelling errors → [retest result].
- Darken the teal chart colour and the red End session button to meet contrast targets → [retest result].
- Remove or rewrite the “Exercise daily.” and “Stay Healthy.” lines in a supportive tone → [retest result].
- Error rate from [A] to [B] in retest; SUS from [x] to [y].

10. Design Rationale, Reflection and Future Work

Major design decisions and trade-offs

- Inclusive metrics over steps: active time and push cycles are more meaningful, but they are estimates that cannot be perfectly accurate, so they are labelled “est.”.
- Flexible goals over fixed targets: reduces pressure, but may make it easier to set goals that are too low; sliders have sensible ranges and encouragement.
- Personal baseline over leaderboards: protects motivation and dignity, but removes the community motivation some users enjoy.
- Manual entry and editing: restores control when sensors miss activity, but adds effort and may reduce data accuracy.

Limitations of the final design

The prototype uses simulated sensor data and was tested only on mobile with a small group. Energy and push-cycle estimates are not clinically validated, only wheelchair rolling is fully designed (hand cycling and strength are placeholders), and the design has not been tested with screen readers or switch controls.

What failed or was redone, and what was learned

The first idea relied on streaks and leaderboards, which research showed would pressure users and ignore differences in ability, so these were dropped for a personal-baseline approach. Early screens also showed

estimates as exact numbers, which was redone with “est.” labels and a sensor notice. The lesson: inclusive design starts by questioning the standard metrics and test early with real users.

Ethical considerations

Health and movement data are sensitive, so the design includes privacy settings and data export, collects only what is needed, and keeps sharing optional. The app avoids medical claims and states that metrics are estimates. Motivational language avoids guilt, shaming and “inspiration” stereotypes about disability. Research participants gave informed consent and all notes were anonymised.

Future improvements

- Real sensor integration and validation of push and energy estimates with users and professionals
- Full support for hand cycling, strength work and other activity types
- Screen reader, switch control and voice support
- Optional, opt-in groups and shared goals with caregivers or coaches
- Tablet and smartwatch versions, and multi-language support

References

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[7] Garmin Ltd., “Wheelchair activity tracking,” Garmin Support. [Online]. Available: https://support.garmin.com

Appendix

- Interview scripts and consent notes
- Raw research notes and survey responses
- Usability test scripts
- Additional screens and sketches